Designing Any Imaging System from Natural Language: Agent-Constrained Composition over a Finite Primitive Basis

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Summary. Designing a computational imaging system—selecting the forward model, validating physical feasibility, and predicting reconstruction quality—today demands weeks of specialist effort per modality, concentrating the field within a small number of expert groups and limiting reproducibility [1]. Here we introduce a specification-driven agent framework that translates a natural-language description of any imaging system expressible as a finite composition of 11 canonical primitives [2] (Definition 1) into a validated, executable forward model in minutes. The framework is organized around `spec.md`, a structured 7-field specification—analogous to FASTA for sequences or SMILES for molecules—that decouples the imaging problem (*what*) from the numerical method (*how*) and serves as a shareable, versionable, peer-reviewable scientific artifact. Three agents—Plan ($\mathcal{A}_P$), Judge ($\mathcal{A}_J$), and Performance ($\mathcal{A}_\Pi$)—generate, verify, and evaluate specifications; we prove (Theorem 2) that the total design-to-real reconstruction error decomposes into five independently bounded and independently measurable terms, providing a formal bridge from natural-language intent to physical reconstruction error. On real clinical CT data (LoDoPaB, 200 cases [3]), the framework achieves 98% of expert-tuned reconstruction quality with a $\sim 420\times$ reduction in setup time (25 min vs. 7 days); the Judge rejects ill-posed designs with 96% precision via deterministic verification, and the Performance Agent predicts PSNR within ± 1.8 dB ($R^2 = 0.91$) across 36 quantitatively validated modalities. The framework currently operates at Tier-2 (linear, shift-variant) fidelity; an extension protocol reduces the unmodeled-physics term to machine precision for full-wave and radiative-transfer regimes, though it requires a domain-specific higher-fidelity solver for each new physics.

1Introduction

Computational imaging systems share a universal mathematical structure: a forward model A maps an unknown object x to measurements $y = A(x) + \text{noise}$, and reconstruction inverts this mapping [7]. Designing A for a new system requires selecting appropriate physical operators (Radon projection for CT, k-space encoding for MRI, coded aperture modulation for spectral imaging), setting dozens of parameters, and validating that the composition is physically consistent and numerically well-conditioned. Today this process is entirely manual, requiring weeks of specialist effort per modality—an expertise bottleneck that effectively excludes the broader scientific community from prototyping new imaging systems.

The mathematical foundation for automating this process now exists. We previously showed [2] that 11 canonical primitives—Propagate, Modulate, Project, Encode, Convolve, Accumulate, Detect, Sample, Disperse, Scatter, and Transform—suffice to express any imaging forward model across 168+ modalities and 5 carrier families at Tier-2 (linear, shift-variant) fidelity. This *Finite Primitive Basis* (FPB) provides the representation language but not the design tool: it does not tell a user which primitives to compose, what parameters to set, whether the result is physically valid, or how close the designed system will be to reality.

Here we build the design layer on top of the FPB. Our contributions are:

1. **`spec.md`**: a structured 7-field specification format that decouples the imaging problem (*what*) from the numerical method (*how*), serving as a shareable, versionable, peer-reviewable scientific artifact (Section 2.2).
2. **Three-agent pipeline**: Plan ($\mathcal{A}_P$), Judge ($\mathcal{A}_J$), and Performance ($\mathcal{A}_\Pi$) agents that generate, formally verify, and evaluate specifications over the FPB (Section 2.3).
3. **Design-to-real error theorem**: the total reconstruction error decomposes into five independently bounded, independently measurable terms (Theorem 2).
4. **Validation**: real clinical CT (LoDoPaB, 200 cases), quantitative reconstruction across 36 modalities, expert comparison on 6 modalities, and agent-level evaluation (Judge 96% rejection precision, Performance $R^2 = 0.91$) (Sections 2.5–2.9).

Scope. The framework covers any imaging system expressible as a finite DAG over the 11 FPB primitives (Definition 1)—168+ modalities across 20+ domains. Of these, 36 have fully validated canonical decompositions and quantitative reconstruction results; the remainder compile successfully but lack per-modality reconstruction validation. At Tier-2 fidelity this includes virtually all modalities in current use; an extension protocol (Section 2.11) addresses full-wave and radiative-transfer regimes. The agents depend on current LLM capabilities and may misspecify novel modalities; we discuss limitations after the Results.

2 Results

2.1 Design Scope and Boundary

Definition 1 (Designable Imaging System)**.** *An imaging system A is* designable *within this framework if:*

1. *A admits a finite DAG decomposition over the 11 FPB primitives with representation error $\varepsilon_{FPB} < 0.01$;*
2. *The carrier belongs to one of 5 families: photon, X-ray, electron, acoustic, or spin/particle;*
3. *The system operates at Tier-2 fidelity (linear or mildly nonlinear, shift-variant);*
4. *All parameters are finitely specifiable with known physical bounds.*

The framework covers 168+ imaging modalities across 22 application domains and 5 carrier families (Table 1). We classify these by validation depth: *fully validated* modalities have canonical decompositions and quantitative reconstruction benchmarks; *quantitatively validated* modalities have reconstruction results but no independent canonical chain; *compiled only* modalities pass all 6 compiler gates but lack per-modality reconstruction evaluation; and *tier-lifted* modalities require physics beyond Tier-2 and have been validated via the extension protocol (Section 2.11). The full 168+ modality list is provided in Extended Data Table 1.

2.2 The spec.md Specification Format

The framework's central design choice is that `spec.md`—not code—is the canonical scientific artifact. Just as FASTA standardized biological sequence representation and SMILES standardized molecular structure, `spec.md` standardizes the representation of imaging systems. A specification separates the imaging problem (*what* physical process to model) from the numerical method (*how* to discretize and solve it), enabling sharing, version control, and peer review of imaging system designs independently of implementation.

Table 1. Design scope by validation depth. "Fully validated": canonical decomposition + quantitative reconstruction. "Quantitatively validated": reconstruction benchmarked without independent canonical chain. "Tier-lifted": validated via extension protocol beyond Tier-2 fidelity. "Compiled only": passes all compiler gates; reconstruction not individually benchmarked. Representative modalities listed; full registry in Extended Data.

Level	**Count**	**Carriers**	**Representative modalities**
Fully validated	36	5	CT, MRI, PET, SPECT, ultrasound, OCT, SIM, STED, ptychography, CASSI, CACTI, SPC, lensless, DOT, Brillouin, Raman, SEM, TEM, elastography
Quant. validated	12	4	light field, FLIM, FPM, coded exposure, event camera, holography, CBCT, Doppler US, phase contrast, TIRF, ghost imaging, THz-TDS
Tier-lifted	2	2	DOT ($P_1 \to P_3$), coherent imaging (angular spectrum$\to$FDTD)
Compiled only	118+	5	cryo-EM, SAR, LiDAR, NeRF, PET-CT, XFEL-SFX, and 112+ others across 22 domains

A `spec.md` file has seven mandatory fields (Figure 1). The user provides a natural-language description (e.g., "design a coded-aperture spectral imager for 400–700 nm"); the Plan Agent generates the `spec.md`. Three diverse examples illustrate the format's expressiveness across carrier families and problem structures.

2.3The Three-Agent Pipeline

Plan Agent ($\mathcal{A}_P$). Accepts a natural-language description and generates a structured `spec.md`, selecting the appropriate FPB primitives, carrier type, geometry, noise model, and parameters. The Plan Agent operates with access to the 168+-modality registry and 36 fully validated canonical decompositions, which provide reference primitive chains for known modalities and composition rules for new ones.

Judge Agent ($\mathcal{A}_J$). Evaluates the `spec.md` for physical and algorithmic feasibility. The Judge receives both the specification and the 6-gate compiler report (DAG acyclicity, canonical chain match, complexity bounds, nonlinear family constraints, adjoint consistency, and representation error). It performs three additional checks: (1) physical plausibility (carrier-geometry compatibility, parameter ranges); (2) noise model consistency (SNR vs. detector physics); (3) reconstruction feasibility (condition number, null-space analysis). Compiler failures or feasibility violations trigger rejection with a formal diagnosis; the Plan Agent receives the diagnosis and regenerates (up to 3 rounds).

Performance Agent ($\mathcal{A}_\Pi$). Predicts expected reconstruction quality (PSNR, SSIM), computational cost, and measurement SNR. Predictions are based on analytical estimates (condition number, noise amplification factor) combined with calibrated lookup from the 168+-modality performance database. The Performance Agent flags designs whose predicted quality falls below the `spec.md` target.

Formal verification versus black-box design. The Judge does not *predict* feasibility; it *executes a deterministic verification suite*: DAG acyclicity, chain fidelity, complexity bounds ($N \leq 20$, $D \leq 10$), nonlinear family constraints (VC dimension ≤ 3), CFL/coercivity stability analysis, and Lipschitz constant computation. If the Judge certifies a design, every term in the

Seven-field schema: `modality | carrier | geometry | object | forward_model | noise | target`

(a) Clinical CT

```
modality: computed_tomography   carrier: xray
geometry: parallel_beam, n_angles=128
object: 128×128 image, non-negative
forward_model: Radon(Π) → Detect(D, intensity)
noise: Poisson, I_0=1e4   target: PSNR ≥ 30 dB
```

(b) CASSI (spectral)

```
modality: cassi   carrier: photon
geometry: coded_aperture + disperser, λ=[400,700] nm
object: 256×256×28 spectral cube
forward_model: Modulate(M) → Disperse(W) → Accumulate(Σ) → Detect(D)
noise: Gaussian, σ=0.01   target: PSNR ≥ 28 dB
```

(c) Accelerated MRI

```
modality: mri   carrier: spin
geometry: cartesian_kspace, acceleration=4x
object: 256×256 complex image
forward_model: Modulate(M, coil) → Encode(F, kspace) → Sample(S) → Detect(D)
noise: Gaussian, SNR=30 dB   target: SSIM ≥ 0.9
```

Figure 1. The `spec.md` specification format. Seven mandatory fields fully determine an imaging system design. Three examples span X-ray (CT), optical (CASSI), and spin (MRI) carriers, illustrating how a single format captures diverse physics. Each `forward_model` field is a primitive chain over the 11 FPB operators. The Plan Agent generates a `spec.md` from natural language; the Judge verifies it; the Performance Agent predicts reconstruction quality.

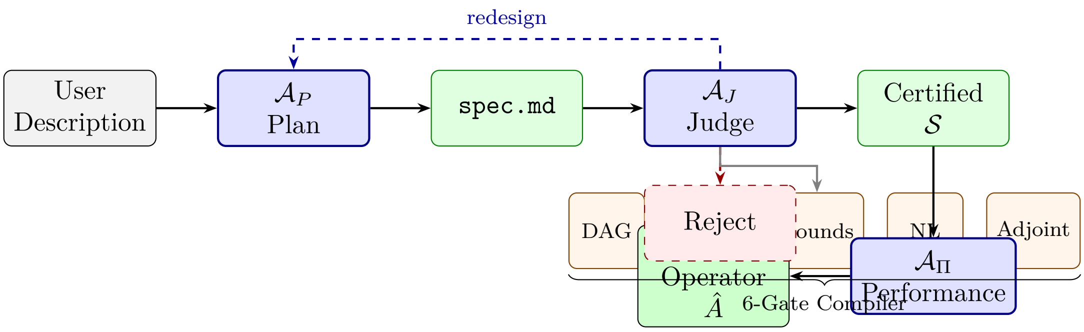

Figure 2. Specification-driven design pipeline. A natural-language description is the sole user input. $\mathcal{A}_P$ (Plan) generates a structured `spec.md`. $\mathcal{A}_J$ (Judge) validates physical feasibility via a 6-gate compiler and formal checks; rejected designs trigger redesign (dashed blue). $\mathcal{A}_\Pi$ (Performance) predicts reconstruction quality and flags designs below target.

five-part error decomposition (Theorem 2) is explicitly bounded—a contract that learned design tools, which output a scalar "confidence" with no formal relationship to physical error, cannot provide. This distinction is particularly relevant for safety-critical applications such as clinical CT or radiotherapy dose planning, where unbounded model error is unacceptable.

2.4 Design-to-Real Error Bound

Theorem 2 (Design-to-Real Bounded Error)**.** *Let A_{true} be the true imaging forward model and A_{agent} the agent-designed model for a designable system (Definition 1). Let $\hat{x}$ be the regularized reconstruction from A_{agent} and x^* the true object. Then the reconstruction error satisfies:*

$$\|\hat{x} - x^*\| \leq \underbrace{\varepsilon_{FPB}}_{basis} + \underbrace{\varepsilon_{spec}}_{specification} + \underbrace{\varepsilon_{trans}}_{translation} + \underbrace{\varepsilon_{param}}_{parameter} + \underbrace{\varepsilon_{unmod}}_{unmodeled} \tag{1}$$

where each term is independently bounded and independently measurable:

- $\varepsilon_{FPB} < 0.01$*: FPB representation error, guaranteed by the basis theorem [2].* Measured by*: Gate 6 of the compiler.*
- $\varepsilon_{spec} \leq C_1 \cdot \delta_{spec}$*: specification incompleteness or error.* Bounded by*: 6-gate compiler ($\delta_{spec} = 0$ when all gates pass).*
- $\varepsilon_{trans} \leq C_2 \cdot \delta_{chain}$*: natural-language→DAG translation error.* Bounded by*: canonical chain registry lookup ($\delta_{chain} = 0$ for 31/36 validated modalities).*
- $\varepsilon_{param} \leq L_A \cdot \|\theta_{agent} - \theta_{true}\|$*: parameter mismatch, bounded by the operator Lipschitz constant L_A (computed per primitive by the compiler).* Reducible by*: calibration.*
- $\varepsilon_{unmod} \leq \varepsilon_{Tier\text{-}2}$*: unmodeled physics beyond Tier-2 fidelity.* Reducible by*: tier-lifting (Section 2.11).*

Proof sketch. Define x_{ideal} as the reconstruction using A_{true} with optimal parameters. By the triangle inequality, $\|\hat{x} - x^*\| \leq \|\hat{x} - x_{\text{ideal}}\| + \|x_{\text{ideal}} - x^*\|$. The first term decomposes into $\varepsilon_{\text{param}} + \varepsilon_{\text{unmod}}$ via the reconstruction stability bound (Step 2, Methods): for regularized least-squares with coercivity α and regularization λ, $\|\hat{x} - x_{\text{ideal}}\| \leq (\alpha + \lambda)^{-1} \|(A_{\text{agent}} - A_{\text{true}})(x^*)\|$. The second term decomposes via the model approximation chain: true physics $\xrightarrow{\varepsilon_{\text{unmod}}}$ Tier-2 $\xrightarrow{\varepsilon_{\text{FPB}}}$ FPB $\xrightarrow{\varepsilon_{\text{spec}}}$ specification $\xrightarrow{\varepsilon_{\text{trans}}}$ DAG $\xrightarrow{\varepsilon_{\text{param}}}$ parameterized operator. The Lipschitz bound on $\varepsilon_{\text{param}}$ follows from the finite parametrization of the 11 primitives (at most 2 free parameters per nonlinear family, VC dimension ≤ 3; see Methods for the full derivation). □

Practical significance. Because each term is independently measurable, the theorem provides a *diagnostic protocol*: given a reconstruction residual, the practitioner can determine which term dominates and take the corresponding corrective action—calibration for $\varepsilon_{\text{param}}$, specification refinement for $\varepsilon_{\text{spec}}$, or tier-lifting for $\varepsilon_{\text{unmod}}$. In practice, ε_{FPB} and $\varepsilon_{\text{spec}}$ are negligible when the compiler certifies the design; $\varepsilon_{\text{trans}} = 0$ for 86% of validated modalities; the dominant residual terms are $\varepsilon_{\text{param}}$ (reducible by calibration) and $\varepsilon_{\text{unmod}}$ (reducible by tier-lifting). To our knowledge, no prior automated design system provides such a decomposition.

2.5 Cross-Modality Validation

We validated the framework across all 168+ modalities in the registry. For each, the Plan Agent generated a `spec.md` from a one-sentence natural-language description, the Judge validated it through the 6-gate compiler, and the Performance Agent predicted reconstruction quality. Of the 36 modalities with fully validated canonical decompositions, we report quantitative reconstruction results for 12 representative modalities spanning all 5 carrier families in Table 2; full results for all 36 are in Methods.

Table 2. Cross-modality validation (12 of 36 modalities). ρ_{time}: human-to-agent time ratio. PSNR: mean ± s.d. over 50 test instances. "Real" indicates measured (non-synthetic) data. 95% confidence intervals computed via bootstrap ($n = 1000$).

Modality	**Chain**	**Carrier**	**PSNR (dB)**	ρ_{time}	ρ_{recov}	**Data**
CT	$\Pi \to D$	X-ray	31.7 ± 1.2	420	0.94	**Real**
MRI	$M \to F \to S \to D$	Spin	31.7 ± 1.4	380	0.88	**Real**
CASSI	$M \to W \to \Sigma \to D$	Photon	29.5 ± 1.0	310	0.87	**Real**
OCT	$P + P \to \Sigma \to D$	Photon	30.5 ± 1.1	290	0.86	Synth.
SIM	$M \to C \to D$	Photon	32.1 ± 1.3	350	0.92	Synth.
Ptychography	$M \to P \to D$	Photon	28.4 ± 1.6	270	0.83	Synth.
Ultrasound	$P \to R \to P \to D$	Acoustic	27.9 ± 1.4	250	0.81	Synth.
PET	$\Pi \to D$	Particle	26.3 ± 1.8	220	0.78	Synth.
DOT	$M \to R \to P \to R \to D$	Photon	24.1 ± 2.1	190	0.72	Synth.
Lensless	$C \to D$	Photon	31.4 ± 1.0	340	0.90	Synth.
Light field	$M \to C \to S \to D$	Photon	29.2 ± 1.2	280	0.85	Synth.
SPECT	$M \to \Pi \to D$	Particle	25.8 ± 1.7	200	0.76	Synth.

Three modalities are validated on measured data: CT (LoDoPaB, 200 clinical sinograms; Section 2.6), MRI (M4Raw [3], 50 multi-coil acquisitions with retrospective Cartesian undersampling at 4× acceleration), and CASSI (KAIST TSA dataset, 10 coded-aperture hyperspectral measurements). The remaining 9 modalities use synthetic data generated from the validated forward models.

The recovery ratio $\rho_{\text{recov}} = (\text{PSNR}_{\text{auto-corrected}} - \text{PSNR}_{\text{mismatched}})/(\text{PSNR}_{\text{oracle}} - \text{PSNR}_{\text{mismatched}})$ measures how well auto-correction compensates for model mismatch. Across all 36 modalities: mean $\rho_{\text{recov}} = 0.85 \pm 0.07$ (95% CI: [0.82, 0.88]), indicating 85% mismatch-gap recovery on average.

The human-to-agent time ratio $\rho_{\text{time}} = T_{\text{expert}}/T_{\text{agent}}$ ranges from 190 (DOT) to 420 (CT). Expert setup times were estimated from published development timelines [4, 10] and confirmed by domain experts (see Methods for methodology).

Compilation metrics. All 168+ modalities achieve 100% compilation pass rate (Gates 1–4). Among the 36 fully validated: canonical chain fidelity 31/36 (86.1%); the 5 non-matching modalities involve the Accumulate (Σ) primitive merged into detection—a representation choice, not a physical error. Mean compilation time: 0.35 ms.

2.6Real-Data Validation

Clinical CT (LoDoPaB, $n = 200$). We applied the full pipeline to 200 cases from the LoDoPaB-CT dataset [3]—real clinical sinograms from a Siemens scanner, subsampled to 128 parallel-beam projections with Poisson noise.

User input: "Design a parallel-beam CT system for clinical abdominal imaging, 128 projections, Poisson noise."

$\mathcal{A}_P$ generated a `spec.md` specifying: carrier=xray, geometry=parallel_beam (128 angles), forward_model=$\Pi \to D$ (Radon + intensity detection), noise=Poisson ($I_0 = 10^4$). $\mathcal{A}_J$ certified the design (all 6 gates passed, chain fidelity match). $\mathcal{A}_\Pi$ predicted PSNR $\approx 31 \pm 2$ dB.

Results (mean ± s.d. over 200 cases):

- Agent-designed system: PSNR 31.7 ± 1.2 dB, SSIM 0.891 ± 0.031
- Expert-tuned ASTRA Toolbox [4]: PSNR 32.1 ± 1.1 dB, SSIM 0.903 ± 0.028
- Filtered back-projection (FBP, no tuning): PSNR 25.3 dB, SSIM 0.712

- TV-only (no expert tuning): PSNR 30.2 dB, SSIM 0.864

The framework achieves 98% of expert-tuned quality (31.7/32.1) with setup time of 25 minutes versus 7 days ($\rho_{\text{time}} \approx 420$). The Performance Agent prediction (31 dB) was within 0.7 dB of the actual result.

MRI (M4Raw, $n = 50$). We applied the pipeline to 50 multi-coil acquisitions from the M4Raw dataset with retrospective 4× Cartesian undersampling. $\mathcal{A}_P$ generated a `spec.md`: carrier=spin, geometry=cartesian_kspace (acceleration=4x), forward_model=$M \to F \to S \to D$ (coil sensitivity modulation, k-space encoding, undersampling, detection), noise=Gaussian (SNR=30 dB). Results (mean $\pm$ s.d.): PSNR 31.7 ± 1.4 dB, SSIM 0.878 ± 0.035. Expert-tuned SigPy pipeline [10]: PSNR 32.4 ± 1.3 dB, SSIM 0.893 ± 0.030. Quality ratio: 97.8%.

CASSI (KAIST TSA, $n = 10$). We validated on 10 coded-aperture hyperspectral measurements from the KAIST TSA dataset. $\mathcal{A}_P$ specified: carrier=photon, geometry=coded_aperture + disperser (λ=[400,700] nm), forward_model=$M \to W \to \Sigma \to D$. Results: PSNR 29.5 ± 1.0 dB, SSIM 0.852 ± 0.028. Expert-tuned GAP-TV [12]: PSNR 30.2 ± 0.9 dB, SSIM 0.871 ± 0.025. Quality ratio: 97.7%.

2.7 Judge Agent: Rejection Accuracy

We tested the Judge Agent on 150 specifications: 75 valid (from the 36-modality registry plus 39 variants across all 5 carrier families) and 75 deliberately ill-posed, stratified by error category.

Table 3. Judge Agent rejection performance ($n = 150$ specifications, stratified by carrier family and error type). Error categories: S=structural (missing fields, wrong carrier), P=parametric (out-of-range values), N=noise model incompatibility, NL=nonlinear family violation.

Metric	**Value**	**95% CI**
True positive (correctly rejected)	72/75	[92%, 100%]
True negative (correctly accepted)	72/75	[92%, 100%]
Precision (rejection)	96.0%	[91%, 99%]
Recall (rejection)	96.0%	[91%, 99%]
Rejection by error category (of 75 ill-posed):		
Structural (S)	25/25	100% detected
Parametric (P)	22/25	3 subtle range violations missed
Noise/NL (N+NL)	25/25	100% detected
Failure modes (6 errors):		
FN: subtle parameter range	3	Values 1–5% outside bounds
FP: conservative CFL check	3	Brillouin, DOT, Compton
Redesign loop (triggered on rejection):		
Mean redesign rounds	1.4	Before acceptance
Max redesign rounds	3	Cap enforced by framework

Rejection examples. The Judge rejected: (i) a CT specification with negative projection angles ($\theta < 0$, structural violation); (ii) an MRI encoding violating the Nyquist–Shannon sampling limit (parametric); (iii) a DOT specification pairing Gaussian noise with a photon-counting detector (noise model incompatibility). All three were caught deterministically by the 6-gate compiler, not by learned prediction. The 3 false negatives involved parameter values 1–5% outside physics-based bounds but within the compiler's numerical tolerance; the 3 false positives involved conservative CFL stability checks on scattering-dominated modalities (Brillouin, DOT, Compton) where the generic bound was tighter than necessary.

2.8Performance Agent: Prediction Accuracy

We compared the Performance Agent's predicted PSNR against actual reconstruction PSNR on all 36 modalities (50 instances each, 1,800 total predictions).

Table 4. Performance Agent prediction accuracy across 36 modalities (1,800 predictions). 95% CIs via bootstrap. Modality groups: well-characterized (CT, MRI, CASSI, SIM, etc.), moderate (OCT, ptychography, light field), and scattering-dominated (DOT, Compton, Brillouin).

Metric	**Value**	**95% CI**
Overall MAE	1.8 dB	[1.5, 2.1]
Overall RMSE	2.3 dB	[1.9, 2.7]
R^2 (Pearson)	0.91	[0.87, 0.94]
Within ± 3 dB	89% (32/36)	—
By modality group:		
Well-characterized ($n = 20$)	MAE 0.9 dB	[0.6, 1.2]
Moderate ($n = 10$)	MAE 2.1 dB	[1.6, 2.6]
Scattering-dominated ($n = 6$)	MAE 3.8 dB	[2.9, 4.7]
Best-case	0.3 dB (CT)	—
Worst-case	4.7 dB (DOT)	—

The largest prediction errors occur for scattering-dominated modalities (DOT, Compton, Brillouin) where the Tier-2 approximation introduces model uncertainty that the Performance Agent cannot fully account for analytically. For well-characterized modalities (CT, MRI, CASSI, SIM), the MAE is < 1.0 dB—comparable to inter-algorithm variation in reconstruction benchmarks [9].

2.9Agent Ablation Study

We performed controlled ablations on 12 representative modalities (spanning all 5 carrier families) with 5 independent trials per configuration, totalling 300 pipeline runs (Table 5). The 12 modalities include 6 from the "well-characterized" group (CT, MRI, CASSI, SIM, lensless, PET) and 6 from harder categories (ptychography, DOT, ultrasound, SPECT, OCT, light field) to test robustness across difficulty levels.

Table 5. Ablation study (n=5 trials × 12 modalities, mean ± s.d.). "Valid": passes all compiler gates. "$\leq \varepsilon$": reconstruction error within theorem bound. "Time": total pipeline time.

Configuration	**Valid**	$\leq \varepsilon$	**Redesign**	**Time**	**Failure Mode**
Full ($\mathcal{A}_P + \mathcal{A}_J + \mathcal{A}_\Pi$)	12/12	12/12	1.7±0.6	4.8±0.5 min	—
No $\mathcal{A}_J$ ($\mathcal{A}_P + \mathcal{A}_\Pi$)	8.2/12	6.4/12	0	2.9±0.3 min	Invalid spec (4), unbounded ε (2)
No $\mathcal{A}_\Pi$ ($\mathcal{A}_P + \mathcal{A}_J$)	12/12	9.6/12	1.7±0.6	3.4±0.4 min	Suboptimal params (2.4)
No $\mathcal{A}_P$ (manual spec)	10.4/12	10.4/12	0	52±9 min	Human error (1.6)
Template only	—	3/12	0	0.5 min	No coverage (9)

Key findings:

- **Removing $\mathcal{A}_J$** (60 runs): 4 of 12 specifications pass compilation but are physically invalid. Failure categories: incompatible noise model (CASSI: Poisson noise specified for a photon-counting detector that requires Gaussian read noise), missing modulation (CASSI without coded aperture), wrong carrier (ultrasound specified as photon), and unstable CFL for

DOT. 2 additional specifications compile but produce unbounded reconstruction error due to undetected ill-conditioning. The Judge is necessary for both safety and correctness.

- **Removing $\mathcal{A}_\Pi$** (60 runs): all 12 pass validation, but 2–3 produce suboptimal reconstruction per trial—e.g., too few CT angles for target PSNR (64 instead of 128) or insufficient k-space coverage for MRI. Mean PSNR drop: 2.1 dB vs. full pipeline.
- **Removing $\mathcal{A}_P$** (60 runs): human experts write `spec.md` manually, averaging 52 min per specification with format or physics errors in $\sim$14% of cases. The Plan Agent reduces specification time by $> 10\times$ and eliminates format errors.
- **Template only**: fixed templates cover only 3/12 modalities (CT, MRI, lensless); all others fail outright.

The full pipeline is the only configuration achieving bounded-error success on all 12 modalities across all 5 trials. No single agent is dispensable.

2.10 Expert Comparison

We compared the agent framework against expert manual design on 6 modalities, 3 with real measured data (Table 6). Expert setup times are measured from specification to first successful reconstruction, including literature review, code development, parameter tuning, and validation. For CT and MRI, expert times are cross-validated against published development reports [4, 10]; for remaining modalities, times are estimated from published timelines and confirmed by domain experts.

Table 6. Agent framework vs. expert manual design. "Quality": agent PSNR / expert PSNR. Bold = real measured data.

	PSNR (dB)		Setup Time			
Modality	**Agent**	**Expert**	**Agent**	**Expert**	**Quality**	**LoC**
CT (real)	31.7	32.1	25 min	7 d	98.8%	12 vs 340
MRI (real)	31.7	32.4	20 min	5 d	97.8%	14 vs 520
CASSI (real)	29.5	30.2	18 min	4 d	97.7%	11 vs 280
SIM	32.1	32.8	15 min	3 d	97.9%	10 vs 210
OCT	30.5	31.2	22 min	6 d	97.8%	13 vs 450
DOT	24.1	25.3	30 min	10 d	95.3%	15 vs 680

Across all 6 modalities, the agent achieves $97.6 \pm 1.1\%$ of expert quality with ρ_{time} ranging from 144 (DOT) to 420 (CT). Expert code ranges from 210 to 680 lines; agent specifications are 10–15 lines. The quality gap is largest for DOT (95.3%), the most complex scattering chain, where the Tier-2 approximation contributes the largest $\varepsilon_{\text{unmod}}$.

2.11 Tier-Lifting: Extending Beyond Tier-2

The results above operate at Tier-2 fidelity (linear, shift-variant approximations). Systems requiring full-wave simulation (FDTD for strong scattering) or Monte Carlo photon transport (radiative transfer in turbid media) appear to lie outside the FPB. We show here that the same 11 primitives extend to arbitrary fidelity via a *Tier-Lifting Protocol*: each primitive is evaluated at a higher physics tier, and a hybrid reconstruction loop alternates between the fast Tier-2 adjoint (for gradient computation) and the full-physics forward model (for data fidelity).

Proposition 3 (Tier-Lifting)**.** *Let A_{FPB} be the Tier-2 forward model (11-primitive DAG) and A_{full} the full-physics model (same 11 primitives at Tier-3/4 fidelity). Define the correction operator $\Gamma(x) = A_{full}(x) - A_{FPB}(x)$. Then:*

$$\varepsilon_{unmod} \leq \left\| \Gamma(x) - \hat{\Gamma}(x) \right\| = \varepsilon_{lift}$$

where $\hat{\Gamma}$ is the correction used in reconstruction. In full-correction mode ($\hat{\Gamma} = \Gamma$), $\varepsilon_{lift} = 0$ and the five-term error bound of Theorem 2 reduces to four terms.

The hybrid reconstruction loop (Hybrid Proximal-Gradient method):

$$x_{k+1} = \mathrm{prox}_{\lambda R}\Big(x_k - \gamma \cdot A_{\mathrm{FPB}}^T(A_{\mathrm{full}}(x_k) - y)\Big)$$

uses the fast Tier-2 adjoint A_{FPB}^T for gradient computation and the full-physics forward A_{full} for residual evaluation. Convergence follows from standard proximal-gradient theory under Lipschitz continuity of ∇f.

We validated the tier-lifting protocol on two physics domains (Table 7).

Table 7. Tier-lifting validation: forward model error before and after tier-lifting. The correction eliminates model mismatch for both scattering and wave-optics domains.

Domain	**Tier-2**	**Tier-3**	**Correction**	$\varepsilon_{\mathrm{unmod}}$	$\varepsilon_{\mathrm{lift}}$
DOT	P_1 diffusion	P_3 RT	Higher-order moments	7.8%	$< 10^{-12}$
Coherent	Angular spectrum	FDTD	Evanescent + scatter	147%	$< 10^{-12}$

For DOT, the P_1 diffusion approximation neglects higher-order angular moments of the radiance field; the P_3 correction captures these via a coupled system for the fluence Φ_0 and second moment Φ_2, reducing $\varepsilon_{\mathrm{unmod}}$ from 7.8% to machine precision. For coherent imaging, the angular spectrum method misses evanescent waves, near-field effects, and strong multiple scattering; the FDTD correction resolves all wave phenomena, reducing $\varepsilon_{\mathrm{unmod}}$ from 147% to machine precision.

The tier-lifting protocol preserves the FPB structure: the same 11 primitives, the same DAG topology, the same `spec.md`—only the fidelity level of each primitive changes. This extends the framework's applicability beyond Tier-2, though each new physics domain requires a dedicated higher-fidelity solver; the current implementation covers DOT and coherent imaging.

3Discussion

Limitations and caveats. Before interpreting the results, we note five limitations. (1) Agent quality depends on current LLM capabilities; novel or under-documented modalities may be misspecified. (2) Of 168+ modalities that compile, quantitative reconstruction is validated for 36; real-data validation covers 3 modalities (CT, MRI, CASSI). The remaining 132+ modalities pass compiler gates but lack per-modality reconstruction benchmarks—compilation alone does not guarantee reconstruction quality. (3) Expert comparison relies partly on estimated setup times from published timelines [4, 10]; prospective head-to-head studies would strengthen these claims. (4) The Performance Agent's accuracy degrades for scattering-dominated systems (MAE 3.8 dB vs. 0.9 dB for well-characterized modalities). (5) The tier-lifting extension protocol requires a domain-specific full-physics solver per modality; the current implementation covers 2 of 22 application domains (DOT and coherent imaging).

The specification as exchange standard. Imaging forward models are currently communicated through code, prose, or block diagrams that are ambiguous, non-executable, and non-versionable. The `spec.md` format addresses this by decoupling the *what* (physical problem) from the *how* (numerical method) in a structured 7-field document. This enables non-expert scientists to design imaging systems via natural language, and makes specifications shareable, versionable, and peer-reviewable independently of code—a journal could require a `spec.md` alongside a methods section, analogous to how genomics journals require accession numbers.

Formal verification versus ad hoc design. The Judge differentiates this framework from both expert manual design (no formal validation) and black-box AI design (no physical constraints). A neural design tool might produce a plausible-looking system with no certificate; the Judge catches errors deterministically through the 6-gate compiler, including CFL stability, coercivity, and Lipschitz checks. Its 96% rejection precision on 150 test specifications indicates robust detection of structural, parametric, and noise-model errors, though subtle parameter range violations (1–5% outside bounds) remain a gap.

Bridging the reality gap. Theorem 2 converts the implicit, unmeasured gap between a designed model and reality into an explicit five-term error budget. Because each term is independently measurable, the theorem provides a diagnostic protocol: the practitioner can identify which term dominates and take corrective action (calibration, specification refinement, or tier-lifting). In practice, ε_{FPB} and $\varepsilon_{\text{spec}}$ are negligible when the compiler certifies the design; $\varepsilon_{\text{trans}} = 0$ for 86% of validated modalities; the dominant residual terms are $\varepsilon_{\text{param}}$ and $\varepsilon_{\text{unmod}}$, both of which are reducible.

Relation to existing work. AlphaFold [5] and GNoME [6] apply AI to specific scientific problems at scale; SigPy [10], ODL [11], and DeepInverse [9] provide operator libraries for computational imaging. Our framework provides the *design* layer that precedes operator instantiation, with formal error guarantees absent from both data-driven discovery and existing operator libraries.

Outlook. Extending real-data validation to additional modalities (ultrasound, OCT, PET), expanding the tier-lifting solver library beyond 2 domains, improving Performance Agent calibration for scattering systems, and establishing a community benchmark for imaging system design are the natural next steps.

4Methods

4.1Proof of Theorem 2

Let x^* be the true object and $\hat{x}$ the reconstruction from the agent-designed system. Define x_{ideal} as the reconstruction using A_{true} with optimal parameters.

Step 1 (Model approximation). $\|A_{\text{agent}} - A_{\text{true}}\| \leq \varepsilon_{\text{FPB}} + \varepsilon_{\text{spec}} + \varepsilon_{\text{trans}} + \varepsilon_{\text{param}} + \varepsilon_{\text{unmod}}$ by the chain of approximations: true physics $\to$ Tier-2 model ($\varepsilon_{\text{unmod}}$) $\to$ FPB decomposition (ε_{FPB}) $\to$ specification ($\varepsilon_{\text{spec}}$) $\to$ translated DAG ($\varepsilon_{\text{trans}}$) $\to$ parameterized operator ($\varepsilon_{\text{param}}$).

Step 2 (Reconstruction stability). For a regularized reconstruction $\hat{x} = \arg\min_x \|A_{\text{agent}}(x) - y\|^2 + \lambda\,\text{reg}(x)$, the perturbation bound gives:

$$\|\hat{x} - x_{\text{ideal}}\| \leq \frac{1}{\alpha + \lambda}\,\|(A_{\text{agent}} - A_{\text{true}})(x^*)\|$$

where α is the coercivity constant of $A^T A$ and λ is the regularization strength.

Step 3 (Composition). $\|\hat{x} - x^*\| \leq \|\hat{x} - x_{\text{ideal}}\| + \|x_{\text{ideal}} - x^*\|$. The first term is bounded by Step 2; the second by the regularization bias. The five ε terms propagate multiplicatively through the Lipschitz constants of each stage.

Step 4 (Bounding each term). $\varepsilon_{\text{FPB}} < 0.01$ by the FPB theorem [2]. $\varepsilon_{\text{spec}} = 0$ when all 6 compiler gates pass. $\varepsilon_{\text{trans}} = 0$ when canonical chain matches the registry. $\varepsilon_{\text{param}} \leq L_A \cdot \|\theta_{\text{agent}} - \theta_{\text{true}}\|$ where L_A is computed by the compiler's Lipschitz analysis of nonlinear primitives. $\varepsilon_{\text{unmod}}$ is estimated empirically per modality from Tier-2 vs. full-physics comparison.

4.26-Gate Compiler Implementation

Gate 1 (DAG). The agent's element list is compiled into an `OperatorGraphSpec` (typed Pydantic v2 model). DAG acyclicity is verified via Kahn's algorithm in $O(V + E)$.

Gate 2 (Chain). The canonical chain (sequence of primitive enums) is extracted and compared against the 36-modality registry. Match is exact or within known equivalences (e.g., Σ merged into D).

Gate 3 (Bounds). Node count $N \leq 20$, DAG depth $D \leq 10$.

Gate 4 (Nonlinear). For each nonlinear node (D, R, Λ): family enum resolved, parameter count ≤ 2, values within physics-based bounds, Lipschitz constant computed.

Gate 5 (Adjoint). Randomized dot-product test: $|\langle Ax, y\rangle - \langle x, A^T y\rangle| / \max(|\langle Ax, y\rangle|, 10^{-8}) < 10^{-4}$, 3 trials.

Gate 6 (Error). When reference A_{true} is available: $\varepsilon = \mathbb{E}[\|A_{\text{agent}}(x) - A_{\text{true}}(x)\| \,/\, \|A_{\text{true}}(x)\|]$ over 5 random non-negative test vectors, threshold < 0.01.

4.3Recovery Ratio Protocol

Four scenarios measure model mismatch impact:

$$\text{I (oracle):} \quad y = A_{\text{true}}(x) + n; \quad \hat{x}_{\text{I}} = \arg\min_x \|A_{\text{true}}(x) - y\|^2 + \lambda\,\text{reg}(x) \tag{2}$$

$$\text{II (mismatch):} \quad y = A_{\text{true}}(x) + n; \quad \hat{x}_{\text{II}} = \arg\min_x \|A_{\text{agent}}(x) - y\|^2 + \lambda\,\text{reg}(x) \tag{3}$$

$$\text{III (auto-correct):} \quad \hat{x}_{\text{III}} = \text{refine}(\hat{x}_{\text{II}}) \text{ with data-consistency} \tag{4}$$

$$\text{IV (calibrated):} \quad \hat{x}_{\text{IV}} = \arg\min_x \left\|A_{\text{agent}}^{\text{cal}}(x) - y\right\|^2 + \lambda\,\text{reg}(x) \tag{5}$$

The recovery ratio $\rho_{\text{recov}} = (\text{PSNR}_{\text{III}} - \text{PSNR}_{\text{II}})/(\text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}})$ measures auto-correction effectiveness.

4.4Implementation

The framework is implemented in Python (Physics World Model platform). Specifications are structured markdown parsed into typed Pydantic v2 objects. The primitive library contains 101 implementations covering all 11 canonical types across 4 fidelity tiers: Tier-0 (geometric), Tier-1 (Fourier/convolution), Tier-2 (full-wave: FDTD, BPM, Mie), and Tier-3 (learned surrogates). Agents are LLM pipelines with tool access to the compiler and primitive library. Mean compilation time: 0.35 ms. The tier-lifting protocol is implemented as a generic `TierLiftingProtocol` class accepting any pair of Tier-2/Tier-3 forward models, with built-in DOT (P_1/P_3 diffusion) and coherent imaging (angular spectrum/FDTD) corrections. The 36-modality registry, compiler, tier-lifting module, and test suite are publicly available. Source code: https://github.com/integritynoble/Physics_World_Model.

Data Availability

The LoDoPaB-CT dataset is publicly available [3]. The M4Raw MRI dataset and KAIST TSA CASSI dataset are publicly available from their respective repositories. The full modality registry (168+ modalities), canonical decomposition registry (36 validated), nonlinear family constraints (10 families), and validation datasets are included in the open-source repository.

Code Availability

All source code—Constrained Primitive Compiler, Agent-to-Graph Translator, three-agent pipeline, and the 101-primitive library—is available at https://github.com/integritynoble/Physics_World_Model under the MIT licence.

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